

LITE MTP: MINIMALIST MULTI-TOKEN PREDICTION FOR EDGE ARCHITECTURES

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Paper under double-blind review

ABSTRACT

Speculative Decoding (SD) exploits the memory-bound nature of Large Language Model (LLM) generation to hide the cost of multiple draft tokens within a single forward pass (Leviathan et al., 2023; Chen et al., 2023). Modern SD techniques such as Medusa and Eagle are designed to maximize operational intensities, so they are highly effective for GPUs, but not Edge devices (Cai et al., 2024; Li et al., 2024). Additionally, they increase the parameter count with their draft-heads, negating the benefits of the small architecture. In this work, we evaluate Medusa’s independent head and Eagle’s sequential feature architectures on Gemma 3 270M (Google DeepMind, 2024), comparing performance between Nvidia A100 GPUs and Apple M-Series Silicon (Apple, 2024). We identify that deep speculation trees optimized for GPUs (Miao et al., 2024) cause a bottleneck shift on edge devices: the overhead of expanded attention masks exceeds the latency savings of reduced memory access, resulting in a $3\text{--}5\times$ slowdown on Apple Silicon (Williams et al., 2009). Conversely, we demonstrate that on-device inference requires a minimalist single-token single-head speculation strategy with compact dense draft heads. Our method, *Lite MTP*, achieves a 7–29% speedup over baseline using the optimized MLX inference engine (Hannun et al., 2023) with under 1% memory overhead.

1 INTRODUCTION

The proliferation of on-device AI has driven the adoption of Small Language Models (SLMs) such as Gemma 3 270M as alternatives or supplements to cloud-based inference for easy workloads. However, adoption is limited by high latency that breaks application interactivity. Furthermore, faster responses expand the frontier of locally viable model sizes. While SLMs have significantly reduced parameter counts, the autoregressive nature of transformer generation is fundamentally memory-bound (Williams et al., 2009). To balance the memory bottleneck and increase speed, Speculative Decoding (SD) with Multi-Token Prediction (MTP) has emerged as a popular paradigm (Leviathan et al., 2023; Chen et al., 2023; Gloeckle et al., 2024). Methods like Medusa’s independent heads (Cai et al., 2024) and Eagle’s sequential feature extraction (Li et al., 2024) increase the operational intensity of the decoding process to better utilize GPUs (e.g., Nvidia A100s). They draft multiple future tokens in a single forward pass to convert a portion of the memory-bottleneck to a compute bottleneck, while being lossless. On GPUs, the cost of these deep multi-token speculation trees (Miao et al., 2024) is effectively hidden due to their massively parallel compute optimized nature. This is in stark contrast to mobile chips because the A100 40GB has an FP16 Machine Balance (ratio of peak FLOPs to peak memory bandwidth) of approximately 201 FLOPs/byte (NVIDIA, 2020), whereas the Apple M4 Pro chip’s GPU only has about 62 FLOPs/byte (Apple, 2024), more than $3\times$ less.

In this work, we revisit MTP for edge inference and find that the best-performing strategy is substantially more minimalist than GPU-oriented speculation. Concretely, we: (i) reproduce strong GPU speedups for Medusa and Eagle on Gemma 3 270M; (ii) show that deep speculation trees can become $3\text{--}5\times$ slower than baseline on Apple Silicon under MLX due to verification overhead from expanded attention masks; and (iii) introduce Lite MTP, a single-candidate, single-head variant that avoids tree attention and adds only 0.3% parameters, yielding 7–29% throughput gains over baseline on Apple M4 Pro.

2 METHODOLOGY

We use Gemma 3 270M as our base SLM because it is a common target for on-device deployment and still has enough breadth to warrant and be capable of speculative decoding. To investigate the

Table 1: Parameter overhead on Gemma 3 270M: dense Medusa heads bloat size (256k vocab), while our draft heads add negligible overhead.

Model Configuration	Head Type	Add. Params	Total Size	Increase
Base Model	—	—	270M	0%
Medusa (Standard, 1 Head)	Dense (d=640)	~171M	441M	+63%
Medusa (Standard, 3 Heads)	Dense (d=640)	~786M	1.05B	+189%
Eagle (Ours, Depth 3)	Fusion + Layer	~6.5M	276.5M	+2.4%
Lite MTP (Ours)	2x Dense (d=640)	~820k	270.8M	+0.3%

hardware-specific performance spectrum, we implement two foundational MTP paradigms (Medusa and Eagle), along with our minimalist Medusa variant (Lite MTP), the lightweight Medusa (Cai et al., 2024) and the heavier Eagle (Li et al., 2024).

2.1 ARCHITECTURE VARIANTS

Medusa Medusa (independent heads) first introduced the MTP speculative decoding paradigm to LLMs. Medusa augments the base model with K independent, lightweight draft heads. Concretely, each head consists of a single linear layer followed by a SiLU activation and an LM-head, predicting future tokens $t + 1, t + 2, \dots, t + k$ conditioned on the backbone’s final hidden state h_t . Once predicted, Medusa relies on a large tree-attention mask (up to 80 nodes) during the verification pass to maximize token acceptance on GPU (Cai et al., 2024; Miao et al., 2024).

Inference Verification At step t , the backbone produces h_t (Leviathan et al., 2023). Heads use this to draft the probability distribution of candidates that we arrange into a greedy tree (Cai et al., 2024). We verify the tree in one backbone pass using tree-attention masks (Miao et al., 2024), and accept only tokens matching the backbone’s greedy choice (Chen et al., 2023). We then repeat the MTP process from the last accepted state (Cai et al., 2024).

Eagle Eagle replaces independent heads with an autoregressive 1-layer transformer over features from early, middle, and later backbone layers, producing depth- k draft trees (Li et al., 2024). This can improve draft coherence, but increases drafting compute, especially for smaller models. Interestingly, Eagle does not have an LM head of its own and uses the base model’s. This was the inspiration for our method to also reuse it.

Ours: Lite MTP For SLMs like Gemma 3 270M, Medusa heads are memory-prohibitive because the vocabulary is large (256k). A single LM head has ~170M parameters. The standard five-Medusa-head setup thus quadruples the model size (Table 1), undermining the point of the SLM. We therefore propose *Lite MTP*, a minimalist Medusa variant that drafts just a single candidate with a compact dense module (two 640×640 layers) and reuses the base model’s LM head for token scoring. The single-candidate prediction allows standard causal attention to be used rather than tree attention, reducing verification overhead that dominates on-device (Section 3.2). We also evaluated both a LoRA adapter on the LM head (rank 64) (Hu et al., 2021) and a no-LoRA design. We found the differences to be small (Table 2), so we adopt the simpler no-LoRA configuration for edge deployment, adding only 0.3% parameters (Table 1).

2.2 TRAINING RIGOR AND LOSS

To ensure draft heads align with the base model’s distribution, we moved beyond generic corpora like FineWeb-EDU (Penedo et al., 2024) and used self-generated trajectories. Specifically, we generated 100k completions by prompting Gemma 3 270M with the OpenHermes 2.5 dataset (Teknum, 2023). This keeps training in-distribution and lets us use KL-based distillation against the backbone. While we initially used Cross-Entropy (CE) loss for simplicity (as in Medusa-1), we found Kullback-Leibler (KL) Divergence consistently improved alignment, matching the backbone’s soft-target distribution (rather than hard labels) increases acceptance rates and downstream throughput.

2.3 EVALUATION SETUP

We measure decoding throughput (tokens per second at batch size 1) on an Nvidia A100 and Apple M4 Pro via MLX (Hannun et al., 2023), sweeping draft tree size (speculative candidates) to capture hardware regime changes. We assume that MLX’s implementation is near-optimal and approximates true hardware performance on Apple Silicon, as it is the standard framework for on-device inference benchmarking. Large trees are empirically effective on GPU, while on-device inference prefers shallow speculation (Figure 1).

Table 2: Recall@k for Medusa head configurations on the self-distilled dataset. KL loss dramatically outperforms CE, while head type differences are minimal.

Loss	LM Head	Head 0		Head 1		Head 2	
		top1	top10	top1	top10	top1	top10
CE	Full	.055	.384	.060	.373	.067	.359
CE	NoLoRA	.053	.341	.052	.312	.056	.300
CE	LoRA-64	.051	.371	.062	.361	.066	.350
KL	NoLoRA	.373	.686	.224	.501	.151	.412
KL	LoRA-64	.419	.752	.258	.580	.182	.487

Table 3: GPU Decoding Throughput on GSM8K (left) and HumanEval (right). Tok/Step is the average total tokens accepted per decoding step (including the 1 token from the backbone). The baseline tok/s are 35.2 and 33.6 for GSM8K and HumanEval respectively. KL divergence training improves Medusa acceptance rates, while Eagle benefits from increased depth and accuracy.

GSM8K				HumanEval			
Method	Config	Tok/Step	Speedup	Method	Config	Tok/Step	Speedup
Baseline	—	1.00	1.00×	Baseline	—	1.00	1.00×
Medusa	2 layers	1.54	1.09×	Medusa	2 layers	1.35	1.01×
Medusa	KL + 2 layers	2.15	1.52×	Medusa	KL + 2 layers	1.96	1.47×
EAGLE	Depth 1	1.94	1.38×	EAGLE	Depth 1	1.94	1.44×
EAGLE	Depth 2	2.78	1.38×	EAGLE	Depth 2	2.89	1.50×
EAGLE	Depth 3	3.42	1.52×	EAGLE	Depth 3	3.68	1.64×

3 EXPERIMENTS & RESULTS

3.1 GPU BASELINE

We first validate that multi-token prediction (MTP) provides meaningful throughput gains for small models in the standard GPU setting. We trained Medusa and Eagle variants on our self-distilled trajectories and evaluated them under decoding throughput (Cai et al., 2024; Li et al., 2024). Across both GSM8K and HumanEval, stronger draft alignment from KL divergence loss correlates with higher acceptance and higher throughput, as switching Medusa to KL-only training increases throughput from 35.9 to 49.5 tps on GSM8K and 33.6 to 49.2 tps on HumanEval (Tables 3–3) (Cobbe et al., 2021; Chen et al., 2021). Eagle shows a clear preference for depth on HumanEval (48.5, 50.5, 56.2 tps at depth 1/2/3), but at 270M scale the heavier drafting module limits how much of its higher token yield translates into end-to-end speed.

Table 4 highlights our central positive result: Lite MTP improves on-device throughput over baseline on Apple M4 Pro by +7.3% on GSM8K and +28.6% on HumanEval at batch size 1. Notably, slightly more aggressive speculation (drafting 3 tokens or verifying at depth 2) does not yield additional gains, reinforcing that the edge-optimal regime prioritizes minimal verification overhead over maximum speculative token yield.

3.2 EDGE FAILURE

We then port the same MTP setup to Apple Silicon using MLX, again measuring decoding throughput (Hannun et al., 2023). While deep-tree configurations excelled on GPU, they are dramatically slower on-edge: tok/s can be 3–5× lower than the MLX baseline. In contrast, Lite MTP’s minimalist single-candidate design remains net-positive (Table 4). Examining profiling in depth reveals that as tree size grows, verification becomes dominated by backbone and LM-head compute, and expanded tree attention makes the backbone pass disproportionately expensive. Concretely, increasing tree size to 80 raises backbone cost by 3.17× on M4 Pro, while it increases only 1.12× on A100 (Apple, 2024; NVIDIA, 2020). In this regime, most time is spent in the backbone pass, so the extra verification compute overwhelms the throughput gains from drafting more candidates.

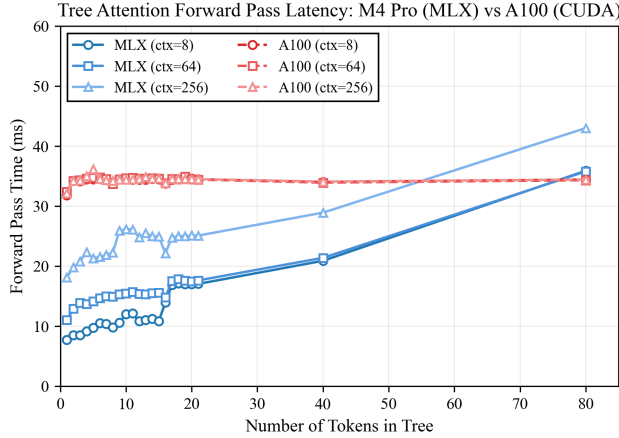


Figure 1: Profiling on MLX (M4 Pro) vs A100: tree size shifts backbone/verification cost.

Table 4: MLX results on GSM8K and HumanEval (batch size 1). Tok/Step = average tokens accepted per decoding step. Lite MTP improves throughput over baseline (+7.3% GSM8K / +28.6% HumanEval).

GSM8K				HumanEval			
Mode	Tok/s	Tok/Step	Speedup	Mode	Tok/s	Tok/Step	Speedup
Baseline	218.4	1.00	1.00×	Baseline	162.1	1.00	1.00×
Lite MTP	234.4	1.39	1.07×	Lite MTP	208.6	1.36	1.29×
Lite MTP-3tok	217.8	1.41	1.00×	Lite MTP-3tok	201.1	1.35	1.24×
Lite MTP-Depth2	215.9	1.48	0.99×	Lite MTP-Depth2	201.1	1.43	1.24×

Table 5: Profiling Comparison (Context = 16). Backbone cost scales 3.17× on MLX vs 1.12× on GPU as tree size grows from 1 to 80.

Tree Size	MLX (M4 Pro)			GPU (A100)		
	Backbone	Verify	Tok/s	Backbone	Verify	Tok/s
1	3.32ms	4.62ms	316.1	31.66ms	31.57ms	18.5
2	3.56ms	5.11ms	299.5	38.49ms	35.67ms	19.0
4	3.66ms	5.70ms	279.2	35.30ms	35.60ms	18.8
8	3.69ms	5.74ms	331.5	35.56ms	35.29ms	21.2
16	7.66ms	10.60ms	250.2	35.56ms	35.37ms	21.4
80	10.53ms	16.29ms	223.7	35.71ms	36.13ms	22.2

3.3 LITE MTP: MINIMALIST SPECULATION

Given the edge failure mode, we use sweeps over tree size (and context) to find the compute-optimal regime on-device. In an idealized sweep, the best tree size for M4 Pro is around 4, but this does not fully capture practical overheads in MLX (e.g., cache behavior when re-arranging KV state and Python overhead for restructuring tree attention). Empirically, tree size 2 is the most robust, our results show that it improves throughput from 218.4 to 234.4 tps (+7.3%) on GSM8K, while larger trees tend to degrade performance (Cobbe et al., 2021). On HumanEval, tree size 2 improves 162.1 to 208.6 tps (+28.6%), with larger trees still positive but weaker (+24%) (Chen et al., 2021). Given the consistent advantage of minimal trees, we present Lite MTP: a hardware-aware recipe for on-device inference using single-token, single-head speculation, where we keep verification overhead small enough to realize net speedups on Apple M4 Pro (Apple, 2024).

4 CONCLUSION

We demonstrate that Multi-Token Prediction strategies optimized for GPUs, achieving up to 1.52× speedup on GSM8K and 1.64× on HumanEval, fail catastrophically on edge devices, resulting in 3–5× slowdowns due to insufficient FLOPs-to-bandwidth ratios to hide deep verification tree overhead. Lite MTP, our hardware-aware solution using single-token single-head speculation with compact dense heads, restores speedups of 1.07× on GSM8K and 1.29× on HumanEval on Apple M4 Pro while adding only 0.3% parameter overhead (820k parameters), compared to 63–189% for

216 standard Medusa heads. This minimal footprint preserves the memory efficiency that makes SLMs
217 viable for on-device deployment. Overall, our results show that for edge inference the simplest form
218 of MTP—single-candidate speculation without tree attention—can outperform the baseline, while
219 deeper GPU-style trees are counterproductive.
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